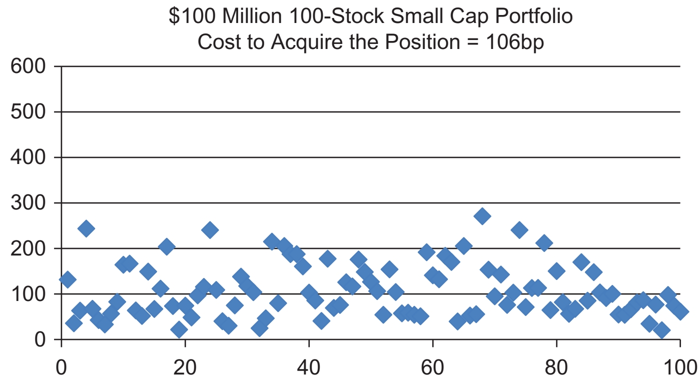

■ **Figure 11.4** Cost to Acquire Each Position in the 100 Stock Portfolio.

up and volatility has spiked. If transactable liquidity is now only half of normal levels and volatility has doubled, the cost to liquidate the position will jump from 106 to 215 bp. Costs are more than 2 times more expensive to sell than to buy. This was caused by market conditions at the time of the sale and not due to any difference in buying or selling sensitivity.

The overall round-trip trading cost of this strategy is approximately 321 bp. If the manager was expecting and planned for a round-trip transaction cost of 200–220 bp they would quickly realize that the liquidation cost of the position eroded their entire incremental alpha and caused them to incur a loss. And the end result is likely that they underperformed their benchmark.

Figure 11.4 shows the market impact cost to acquire each of the 100 stocks in our example. Notice the extent that this cost can vary across all names in the portfolio. The range of cost is from 21 bp (least expensive) to 271 bp (most expensive). The actual cost to trade is determined from the dollar allocation to each name as well as the stock's liquidity (ADV) and volatility. And as this analysis shows—actual trading cost by stock can vary tremendously.

Figure 11.5 shows the market impact cost to liquidate each of the 100 stocks in our example. The stock by stock cost in this example varies from 45 to 527 bp with an average of 215 bp. Again, notice how much higher the liquidation cost is compared to the acquisition cost shown in Figure 11.4.

Formulaically, the liquidation cost in a stressed trading environment computed directly from the simplified I-Star market impact model is as follows.